

Data Quality for Derived Preference Graphs: Construction Sensitivity and Repair Outcomes in Multi-Ranker Retrieval

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Preference graphs built from multiple rankers are derived data artifacts, not ground truth: their edges, weights, and cycles depend on how heterogeneous scores are normalized, thresholded, and aggregated before any repair algorithm is applied. We formalize that construction problem as a seven-dimension data-quality audit covering provenance, calibration, vote semantics, conflict structure, repair quality, downstream utility, and reproducibility. We study four retrieval benchmarks (SciDocs, FiQA, HotpotQA, and BRIGHT) with stored BM25, TF-IDF, and MiniLM scores. Raw score margins are severely scale dominated: BM25's conditional mean edge-weight share is 0.988 under the raw construction but 0.512 after per-query, per-ranker normalization. That change materially alters retained votes, cyclicity, removed edges, and several repaired-versus-unrepaired signs. Under the primary normalized protocol, repair remains structurally active, but no repaired-versus-unrepaired nDCG cell survives Holm correction in the canonical design, the larger-pool $P > k$ study, or direct exact-repair checks. Larger pools confirm that repair can change top- k membership, yet those membership changes still do not yield a reliable aggregate improvement. Exact SCIP repair improves the graph objective but does not reveal a retrieval gain that greedy repair missed. The main contribution is therefore methodological rather than algorithmic: claims about graph repair must report normalization, vote construction, pooling, and evaluation cutoffs as first-class data-quality choices, because those choices determine both the graph being repaired and the effects evaluation can observe.

CCS Concepts: • **Information systems** → **Data cleaning**; **Data quality**; **Retrieval models and ranking**.

Additional Key Words and Phrases: preference graphs, data quality, rank aggregation, score normalization, feedback arc set repair, retrieval evaluation

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1 INTRODUCTION

Preference graphs are attractive because they preserve disagreement among retrieval systems rather than collapsing every score into a single fused number. For a query-specific candidate set, a directed edge $u \rightarrow v$ records support for ranking document u ahead of v , while the edge weight records how strongly that support should count. That representation is useful only if the graph itself is trustworthy. When a graph is built from heterogeneous BM25, TF-IDF, and neural similarity scores, its structure depends on score comparability, tie handling, vote thresholds, and candidate-pool design before any repair step sees it. The graph is therefore a derived data artifact whose quality must be audited, not assumed.

This paper studies that audit problem in the concrete setting of feedback-arc-set repair for multi-ranker retrieval. Repair is often treated as if a better graph-internal solution should translate into a better retrieval ranking, yet the optimization target is structural consistency, whereas retrieval metrics evaluate relevance against qrels. We examine the construction pipeline that connects those

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two objects. Across SciDocs, FiQA, HotpotQA, and BRIGHT, we measure how normalization, vote semantics, candidate pooling, and the distinction between pool size P and metric cutoff k change the preference graph and the apparent effect of repair.

The paper is not a new ranking algorithm. Its contribution is a data-quality framing and a measured negative result: structural repair remains active, but reliable repaired-versus-unrepaired retrieval gains do not survive once construction choices, paired testing, multiplicity correction, larger-pool evaluation, and exact repair are all checked explicitly.

The main contributions are:

- We define the query-specific preference graph as a derived data-quality object and organize its failure modes into a compact, operational seven-dimension audit taxonomy.
- We show that raw heterogeneous score margins are not a neutral starting point: unnormalized construction is scale dominated, and normalization materially changes retained votes, cyclicity, and removed edges.
- We separate structural repair activity from downstream retrieval utility. Repair changes the graph and, under $P > k$, can change top- k membership, but no repaired-versus-unrepaired nDCG family in this study survives Holm correction.
- We show that the null retrieval conclusion is not a greedy-repair artifact, a candidate-pool artifact, or a baseline-coverage artifact: it persists under exact SCIP repair, larger pools, alternative pool policies, and added graph-ranking baselines.
- We distill the results into a practitioner-facing audit logic for future preference-graph studies and deployments.

2 BACKGROUND AND RELATED WORK

Preference-graph methods sit between rank aggregation, probabilistic pairwise ranking, fusion, and data-quality auditing. Rank-aggregation and feedback-arc-set literatures study how to extract consistent orderings from conflicting pairwise evidence [1, 6, 8, 9]. Probabilistic pairwise models such as Bradley–Terry and Rank Centrality consume the same kind of graph but optimize different principles over it [2, 13]. In all of those cases, the graph is usually treated as a valid input rather than as an object with its own measurable quality.

Information retrieval already has graph-free ways to combine heterogeneous rankers, notably RRF, CombSUM, and Borda-style fusion [5, 7, 10]. That literature also makes the score-comparability problem explicit. Hybrid lexical-semantic fusion is sensitive to score normalization, and one reason rank-based methods remain attractive is that they avoid comparing incompatible raw magnitudes directly [3, 12]. The same issue becomes more consequential in a weighted preference graph, where raw margins determine both edge presence and edge weight.

Our framing is closest to data-quality work on derived analytical artifacts. Intrinsic properties of the graph, such as cyclicity or removed feedback-arc weight, can diverge from contextual task utility, just as intrinsic and contextual data quality can diverge more generally [20]. This derived-artifact perspective is consistent with work on data cascades and benchmark error propagation, which shows that upstream curation choices silently shape downstream conclusions [14, 16]. The paper operationalizes that perspective for preference graphs rather than borrowing a general framework and applying it loosely.

Finally, recent LLM-as-judge and pairwise-ranking studies reinforce the same measurement lesson: parser defaults, prompt order, and positional biases can manufacture edges that are then misread as meaningful inconsistency [15, 17, 19, 22, 23]. We use stored LLM judgments only as a bounded protocol-quality check, not as primary evidence, but they show that the construction-first logic is not limited to classical rankers.

Table 1. Compact audit taxonomy for derived preference graphs.

Dim.	Failure mode	Diagnostic used here
A	hidden coverage or abstention asymmetry	candidate-pool overlap, missing-score abstention, qrels-pair eligibility
B	one ranker dominates edge weight by score scale	conditional ranker weight share, raw-vs-normalized edge and sign changes
C	vote semantics manufacture edges	tie abstention, parser-default and position-bias audit for stored LLM judgments
D	cyclicity is over-interpreted	pre/post mutual-pair deletion, mutual-pair attribution, residual cyclicity
E	repair quality is assumed from activity alone	removed weight, greedy-versus-exact comparison, exact optimality status
F	graph repair is assumed to improve retrieval	paired nDCG tests, larger-pool $P > k$ evaluation, baseline comparison, influence checks
G	claims are not reproducible	manifests, fixed stored inputs, claim-to-evidence audit, released scripts

3 AUDIT FRAMEWORK AND METHODOLOGY

For a query q with candidate set D_q and rankers $s \in \{\text{BM25, TF-IDF, MiniLM}\}$, the audited object is the weighted directed graph $G_q = (V_q, E_q, w_q)$ with $V_q = D_q$. It is the output of six named stages: candidate-pool construction, score normalization, pairwise vote extraction, abstention handling, thresholding, and edge aggregation. The graph is not treated as latent truth; it is the product of those stages and inherits their failures. Figure 1 summarizes the audited pipeline.

Our primary protocol uses per-query, per-ranker min-max normalization:

$$\bar{x}_{q,s}(d) = \frac{x_{q,s}(d) - \min_{d' \in D_q} x_{q,s}(d')}{\max_{d' \in D_q} x_{q,s}(d') - \min_{d' \in D_q} x_{q,s}(d')}. \quad (1)$$

Votes are then extracted from pairwise margins between normalized scores, with ties and missing scores treated as abstentions rather than imputed preferences. Aggregated edge weights are thresholded within one named protocol/regime cell. The primary paper reports only the normalized retention-matched protocol in full; raw margins and additional threshold families are reported as robustness checks in the supplement.

Repair removes a minimum-weight feedback arc set:

$$F_q^* = \arg \min_{F \subseteq E_q} \sum_{e \in F} w_q(e) \quad \text{s.t.} \quad (V_q, E_q \setminus F) \text{ is acyclic.} \quad (2)$$

Primary results use greedy cycle peeling; exact SCIP solves are used as a robustness check. From the repaired or unrepaired graph, we extract graph scores and, for hybrid methods, combine them with normalized prior scores:

$$h_q(d) = (1 - \alpha) \hat{s}_q^{\text{prior}}(d) + \alpha \hat{s}_q^{\text{graph}}(d), \quad (3)$$

with $\alpha = 0.3$ in the primary analysis and a small sweep reported in Supplement Section B.

Table 1 turns the construction problem into a compact audit taxonomy. The key decision rule is that cyclic or contradictory structure is called a construction artifact only when a named upstream choice is shown to create it, not merely when the pattern looks suspicious. Raw-score scale dominance and parser-default LLM edges are examples of measured artifacts. Residual conflict after those controls is reported as residual conflict, not automatically as genuine disagreement.

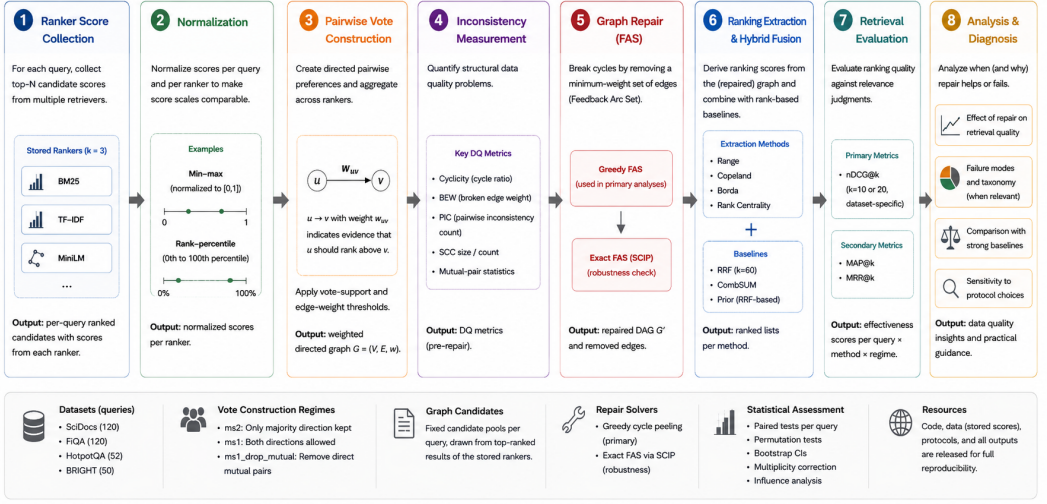


Fig. 1. Pipeline and audit schematic. Stored ranker scores define a shared candidate pool, are normalized per query and ranker, converted into pairwise votes, aggregated into a weighted preference graph, optionally repaired, and then evaluated against qrels. The data-quality audit checks the graph at each stage rather than treating it as a fixed input to repair.

4 EXPERIMENTAL DESIGN

We use four public benchmarks: SciDocs, FiQA, HotpotQA, and BRIGHT [4, 11, 18, 21]. The pipeline reuses stored BM25, TF-IDF, and MiniLM scores and fixed stored query-ID lists. It does not regenerate upstream retrieval. Queries are filtered only for qrels eligibility; this excludes 18 of HotpotQA’s 70 stored IDs and leaves all other stored IDs usable. Candidate pools are formed from the union of documents scored by at least one ranker and, when needed, truncated by an RRF-centered pool policy in the primary design. Alternative pool policies are checked in Supplement Section D.

The original design evaluates the full within-pool order at $(P, k) = (20, 20)$ for SciDocs, FiQA, and BRIGHT and $(10, 10)$ for HotpotQA. To answer whether repair changes membership when the evaluated prefix is smaller than the available pool, we also run a larger-pool study with verified common stored depths of 50 for SciDocs, FiQA, and BRIGHT and 35 for HotpotQA.

Primary confirmatory inference uses paired query-level nDCG deltas with Holm correction. Bootstrap intervals, leave-one-out influence analysis, exact repair, baseline expansion, alternative pools, and independent protocol families are robustness checks rather than separate headline families. MAP and MRR are secondary diagnostics only. Equivalence testing is reported narrowly to avoid over-reading non-significance.

The compared methods include graph-free baselines (Prior, RRF, CombSUM, and Borda), graph-only rankings (Copeland, balance, Markov, PageRank, RankCentrality, and Bradley–Terry), and graph-plus-prior hybrids; greedy repair is the primary intervention and exact SCIP repair is the validation check. Table 2 summarizes the fixed dataset-specific depths and evaluated cells.

5 RESULTS

5.1 Construction Quality Dominates the Graph

The first finding is that raw weighted construction is not a benign baseline. Under the raw-margin ablation, BM25 contributes a mean conditional edge-weight share of 0.988 whenever it participates

Table 2. Datasets and prespecified evaluation settings.

Dataset	Stored IDs	Usable	Depth	Evaluated cells
SciDocs	120	120	50	canonical (20, 20); larger-pool (20, 5), (20, 10), (20, 20), (50, 5), (50, 10), (50, 20)
FiQA	120	120	50	same as SciDocs
HotpotQA	70	52	35	canonical (10, 10); larger-pool (10, 5), (35, 5), (35, 10), (35, 20)
BRIGHT	50	50	50	same as SciDocs

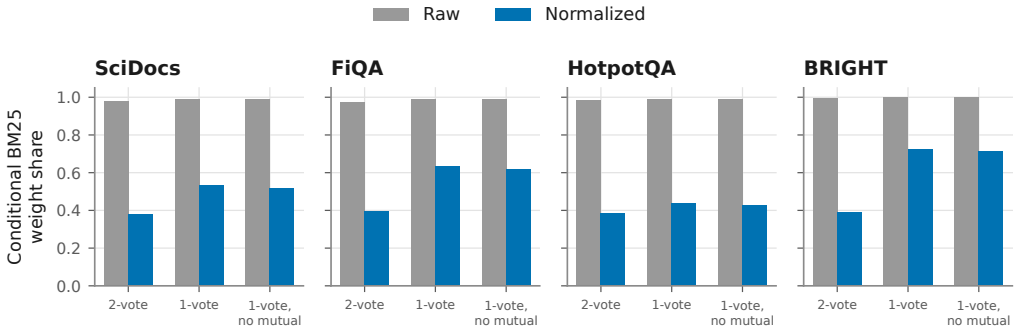


Fig. 2. Conditional BM25 edge-weight share under the raw-margin ablation versus the primary normalized protocol. Raw construction is near-total BM25 domination in every panel; normalization restores a genuinely mixed graph.

in an edge. Under the primary normalized protocol, that mean falls to 0.512. Figure 2 shows that this is consistent across all four datasets and all three vote regimes. The graph built from raw margins is therefore largely a single-ranker graph with multi-ranker labels attached.

Normalization changes more than scale. It changes retained votes and removed edges, and several repaired-versus-unrepaired signs flip between raw and normalized construction. Under the normalized primary protocol, the active ms1 regime is highly cyclic before deleting direct mutual pairs, but much of that cyclicity is not longer-cycle nontransitivity. Figure 3 shows the decomposition. HotpotQA drops from 63.5% cyclic queries to 1.9% after mutual-pair deletion, whereas FiQA retains a larger residual (98.3% $\rightarrow$ 30.8%). The correct report is therefore two numbers: headline cyclicity and post-mutual residual cyclicity.

5.2 Repair Changes Structure More Reliably Than Retrieval

Repair is structurally active under the normalized ms1 graphs. Normalized removed weight ranges from 0.029 to 0.080 of total graph weight across datasets, while ms2 is acyclic by construction. What does not follow is a reliable retrieval benefit. The active canonical family contains 20 ms1 nDCG cells; the full canonical family contains 60 cells across all three regimes. None survives Holm correction. Figure 4 shows why: some point estimates are positive, but most paired intervals still cross zero. Table 3 condenses the main empirical answers.

The larger-pool study resolves the obvious $P = k$ objection. When $P > k$, repair does change the top- k set: the mean top- k membership-change rate rises to 10.6% compared with 0% in the original $P = k$ design. Yet the retrieval conclusion does not reverse. None of the 110 active larger-pool ms1 cells survives Holm correction, and the smallest raw p-value still inflates to a non-significant

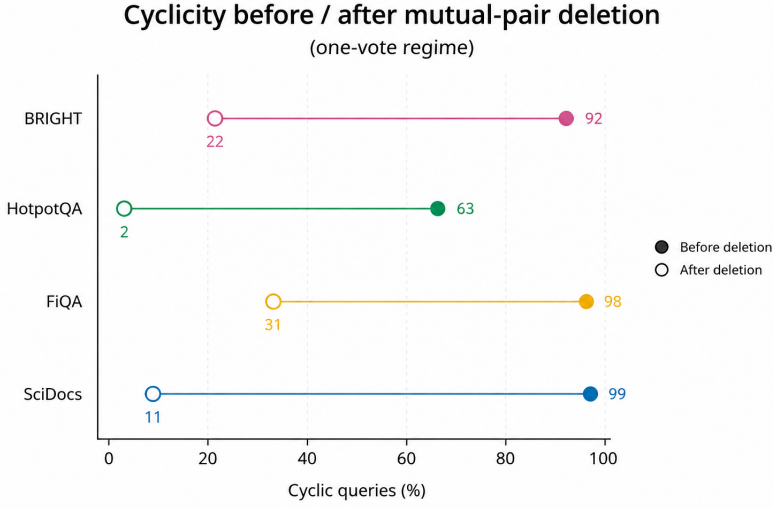


Fig. 3. Cyclic-query percentage under ms1 before and after deleting direct mutual pairs. Most active-regime cyclicity is direct contradiction, not longer-cycle structure, but the residual is still nonzero and dataset dependent.

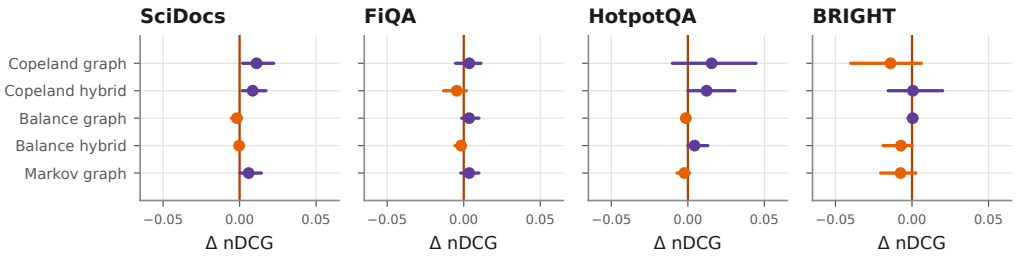


Fig. 4. Repaired-minus-unrepaired $\Delta nDCG$ with paired 95% bootstrap intervals for the active ms1 regime. Some point estimates are positive, but the family as a whole does not survive Holm correction. Zero-effect ms2 and ms1_drop_mutual rows are omitted for readability.

adjusted decision. Repair can therefore change who enters the evaluated prefix without delivering a reliable aggregate nDCG gain.

5.3 The Null Result Persists Under Exact Repair and Robustness Checks

The absence of reliable positive evidence is not a greedy-repair artifact. Exact SCIP repair reaches proven optimality on every nonempty graph checked (1,025/1,025) and removes less total edge weight than greedy, yet the retrieval decision does not change: 0/36 canonical exact cells and 0/56 larger-pool exact cells survive Holm correction. The paper's core contrast is therefore sharp: improving the graph objective does not imply improving retrieval.

Table 3. Primary empirical findings retained in the main paper.

Question	Main evidence
Does raw construction mix evidence fairly?	No. Conditional BM25 edge-weight share is 0.988 raw versus 0.512 normalized.
What does active cyclicity mainly represent?	Mostly direct mutual contradiction. Under ms1, cyclicity drops from 63.5–99.2% to 1.9–30.8% after mutual-pair deletion.
Is repair structurally active?	Yes. Normalized ms1 repair removes 0.029–0.080 of total graph weight; ms2 removes 0.
Does repair reliably improve canonical nDCG?	No. Holm rejects 0/20 active canonical ms1 cells and 0/60 full canonical cells.
Does larger-pool evaluation rescue the claim?	No. Repair changes top- k membership when $P > k$ (10.6% mean rate versus 0% at $P = k$), but Holm still rejects 0/110 active larger-pool cells.
Do exact repair or extra base-lines change the conclusion?	No. Exact SCIP repair is proven optimal on 1,025/1,025 nonempty graphs yet yields 0/36 canonical and 0/56 larger-pool Holm-significant cells; CombSUM and RRF remain competitive.

Table 4. Compact robustness and interpretation summary.

Check	Result
Exact repair quality	SCIP solved 1,025/1,025 nonempty graphs to proven optimality; retrieval remains 0/36 canonical and 0/56 larger-pool Holm-significant.
Independent protocol families	Joint families across alternative normalization and threshold definitions remain null after Holm/BH correction.
Candidate-pool robustness	Alternative pools materially change eligible documents and removed edges, but not the corrected retrieval conclusion.
Baseline fairness	CombSUM and RRF remain competitive; no evidence that the RRF-centered primary pool creates a false repaired-graph advantage.
Power / MDE	Median observed absolute delta 0.0036 versus Holm-adjusted median 80% MDE 0.0207 in the active larger-pool family.
Equivalence	Established only for a minority of cells: 13/110 at ± 0.005 and 32/110 at ± 0.010 .

The conclusion is also not limited to one convenient construction. Joint Holm/BH families over independently-defined normalization protocols, alternative pool policies, and added graph-ranking baselines remain null. Simple baselines stay strong: under the primary ms1 protocol, CombSUM has the best dataset-macro mean nDCG (0.554), RRF is close (0.546), and the best repaired Copeland hybrid is effectively tied with RRF (0.546). The manuscript therefore supports graph repair as a measurable intervention, not as a new default fusion winner.

Power analysis clarifies what the null does and does not mean. In the active larger-pool family, the median absolute observed repaired-minus- unrepaired mean delta is 0.0036, whereas the median analytic 80% MDE is 0.0133 at nominal $\alpha = 0.05$ and 0.0207 after Holm correction. Larger effects are ruled out more confidently than very small ones. Equivalence results are correspondingly limited: only 13/110 cells at margin ± 0.005 and 32/110 cells at margin ± 0.010 reject non-equivalence after Holm correction. The right conclusion is no reliable positive evidence here, not blanket practical equivalence. Table 4 collects the compact robustness outcomes.

6 DISCUSSION

The results support a narrower and stronger conclusion than “repair helps” or “repair never helps.” Preference-graph repair is inseparable from graph construction. Normalization, vote semantics,

pooling, and evaluation cutoff jointly determine the graph being repaired and the effects the evaluation can see. In this study, raw margins create a scale-dominated graph; normalized construction removes much of that artifact; repair remains structurally active; and the downstream retrieval claim still fails.

Three practical implications follow. First, normalization is part of the model, not preprocessing trivia. A weighted preference graph built from heterogeneous raw margins can encode score scale rather than evidence. Second, repair quality and downstream utility must be reported separately. Exact SCIP repair improves the structural objective but does not reveal a retrieval gain that greedy repair missed. Third, top- k membership change is a necessary but insufficient condition for a useful repair effect. The $P > k$ study shows that observing membership changes does not by itself turn the null into a positive result.

For practitioners, the audit logic is straightforward:

- name the candidate-pool policy and check at least one alternative;
- normalize heterogeneous scores before interpreting weighted votes;
- report mutual-pair and longer-cycle cyclicity separately;
- test retrieval claims with paired, corrected inference rather than inferring them from structural activity alone.

Stored LLM judgments illustrate the same point in a different modality: parser defaults and position bias can manufacture edges, so any future LLM-native preference graph should be audited as a construction pipeline before its inconsistency is interpreted.

7 LIMITATIONS

The remaining boundaries are substantive, not already-resolved reviewer concerns.

Finite power for small effects. The study is well positioned to rule out larger repaired-versus-unrepaired gains, but many smaller effects remain below the Holm-adjusted detectable scale.

Ranker and benchmark scope. The evidence is limited to four benchmarks, three stored rankers, and the tested graph-construction choices. The paper does not claim transfer to learned fusion, cross-encoder reranking, or online evaluation.

Candidate-pool composition remains structurally sensitive. Pool choice changes which documents and edges are present even though the null retrieval conclusion is robust across the pool policies tested.

LLM evidence is intentionally bounded. Stored LLM judgments are used only for protocol-quality auditing and do not independently validate the mechanical normalized study.

External validity is limited to tested graph-construction regimes. The paper does not claim that every preference-graph repair setting should behave identically to the four benchmarks studied here.

8 CONCLUSION

Preference graphs in multi-ranker retrieval are derived data artifacts. The main result of this paper is that their construction quality matters at least as much as the repair algorithm applied afterward. Raw heterogeneous score margins create a scale-dominated graph; normalization materially changes retained votes, cyclicity, and repair decisions; repair then improves structural consistency without yielding a reliable aggregate retrieval gain under the tested settings.

That negative result is informative because it survives the checks most likely to overturn it: larger pools with genuine $P > k$ separation, exact SCIP repair, alternative pools, added baselines, and power-aware interpretation. The contribution is therefore a reusable audit logic for derived

preference graphs, together with evidence that structural graph improvement and downstream retrieval utility should be treated as separate quality dimensions.

9 DATA AVAILABILITY AND REPRODUCIBILITY

Data availability. SciDocs, FiQA, HotpotQA, and BRIGHT are public benchmarks and are cited from their original sources [4, 11, 18, 21]. This paper redistributes no raw third-party document collection.

Artifact availability. The anonymous review artifact includes the manuscript source, supplement, stored aggregate tables, protocol manifests, hashes of score files and qrels, and the scripts used to regenerate the reported figures and statistics. The primary evidence set is the normalized retention-matched package; raw margins are retained only as an ablation.

Reproducibility. All numerical claims in the main paper trace to stored intermediates, fixed query-ID lists, released scripts, and machine-readable manifests. Upstream retrieval and paid APIs do not need to be rerun for the mechanical regeneration reported here.

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